



CMIP5 model evaluation for extreme ocean wave height responses to ENSO

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Abstract

The El Niño–Southern Oscillation (ENSO) exerts significant influences on extreme significant wave height (SWH) but climate model capabilities in reproducing the observed ENSO impact on SWH have not been evaluated. This study assesses the performances of the Coupled Model Intercomparison Project phase 5 (CMIP5) models in term of extreme SWH responses to ENSO over the Indo-Pacific Ocean focusing on December–February (DJF). 18 CMIP5 models are evaluated using their historical simulations for 1950–2005 in view of the ERA-20C reanalysis. A non-stationary generalized extreme value (GEV) analysis is employed to fit DJF maxima of 6-hourly SWHs and obtain the extreme SWH response patterns to ENSO by incorporating Niño3.4 index as a covariate. Results show that CMIP5 models can on average capture the major observed mean and extreme SWH responses to ENSO, including the increased SWH over the northeastern North Pacific (NENP) and the decreased SWH over the Maritime Continent (MC) during El Niño. The inter-model relations between ENSO characteristics and SWH responses are further examined for the two hotspot regions (NENP and MC). It is found that ENSO intensity is a dominant factor determining simulated SWH over the NENP such that models with stronger ENSO simulate stronger SWH responses. In contrast, for the MC, the sea level pressure teleconnection pattern significantly affects the inter-model spread in SWH responses, also explaining the systematic underestimation of SWH responses over the region. Implication is that ENSO intensity and atmospheric teleconnection patterns need to be considered for better simulations and reliable predictions of extreme SWH variability.

Keywords Extreme wave heights · ENSO · CMIP5 · Model evaluation · Teleconnection pattern

1 Introduction

Extreme ocean wave is an essential component of global climate system due to its devastating impact on coastal communities, infrastructures, and environment. Thus, a better understanding of the observed variability and change in extreme ocean waves is crucial to prevent associated social and economic disasters. Basically, extreme ocean wave height is related to strong storms which bring the high

surface wind speed (Young 1999; Wang et al. 2010; Young et al. 2012). For example, coastlines of US and East Asia are heavily affected by several hurricanes and typhoons annually, which accompany extreme wave heights (Bromirski and Kossin 2008; Hemer et al. 2011; Park et al. 2014).

Previous studies analyzed the observed trends in significant wave heights (SWH) over the North Atlantic (NA) and North Pacific (NP) Ocean (e.g., Wang and Swail 2001; Wang et al. 2006, 2012; Semedo et al. 2011). They found a significant increase in extreme SWH over the northeast NA in winters, the central NP in boreal winter and spring (April–June). Larger increases in extreme SWH were observed in the Southern Ocean (Young et al. 2011, 2012; Young and Ribal 2019; Meucci et al. 2020), including the extra-tropical South Indian Ocean (Sreelakshmi and Bhaskaran 2020). Examining future projections of extreme SWH using multiple climate models, other studies found that extreme SWH will increase over the tropics (mainly in eastern tropical Pacific) and Southern high latitudes in the

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future (Dobrynin et al. 2012; Fan et al. 2013, 2014; Hemer et al. 2013; Semedo et al. 2013; Wang et al. 2014; Casas-Prat et al. 2018; Kaur et al. 2021). In this regard, recently, the Coordinated Ocean Wave Climate Project (COWCLIP; Hemer et al. 2010b, 2012) developed the multi-method ensemble projections of future changes in global wave climate by compiling the numerous independent studies (Hemer et al. 2013; Morim et al. 2019, 2020; Krishnan et al. 2021).

In addition to the long-term trends, the extreme SWH fluctuates on interannual-to-decadal timescales, depending on regions and seasons (Wang and Swail 2001; Hemer et al. 2010a; Kumar et al. 2016). Previous studies have found that various modes of natural climate variabilities are responsible for the large-scale variability of extreme wave heights (e.g., Izaguirre et al. 2011; Kumar et al. 2016, 2019; Gupta et al. 2017; Marshall et al. 2018; Patra et al. 2020), including El-Niño Southern Oscillation (ENSO), Southern Annular Mode (SAM), Indian Ocean Dipole (IOD), North Atlantic Oscillation (NAO), North Pacific Oscillation (NPO), and Pacific Decadal Oscillation (PDO). ENSO and PDO affect extreme SWH over the northeast NP during DJF while strongest responses to the NAO are evident throughout northern NA (Kumar et al. 2016; Patra et al. 2020). Recently, Patra et al. (2020) reported that extreme SWH around the northeast NP in DJF can be affected by the NPO-ENSO combinations. The IOD impact on extreme SWH is strong over the eastern tropical and southern Indian Ocean during boreal autumn (Kumar et al. 2019). The seasonal variations in SAM affect extreme SWH at various Southern mid- and high latitudes with varying zonal extents (Marshall et al. 2018; Patra et al. 2020). The SAM not only exerts the significant impact over the Southern Ocean year-around, but also drives changes to northern and tropical oceans (Marshall et al. 2018; Wandres et al. 2018; Patra et al. 2020).

The ENSO is a dominant mode of climate variability that arises from air-sea interaction over the equatorial Pacific. On timescale ranging from about two to seven years, the equatorial Pacific sea surface temperature (SST) oscillates between anomalously warm (El Niño) and cold (La Niña) conditions, having widespread effects on the global climate through atmospheric teleconnections (Collins et al. 2010; Stevenson 2012; Yeh et al. 2018). Previous observation-based studies reported that by modulating atmospheric circulation patterns and associated surface wind speed over many regions, ENSO exerts significant influences on the interannual variability of extreme SWH over Pacific and Indian Oceans (Izaguirre et al. 2011; Kumar et al. 2016), particularly inducing increased SWHs in the northeast NP and decreased SWHs over Maritime Continent (MC) during DJF (Menendez et al. 2008; Kumar et al. 2016; Patra et al. 2020) and the Indian Ocean subregions during June–August (Kumar et al. 2016, 2019; Remya et al. 2020).

Global climate models (GCMs) are the most useful tool to enhance our understanding of the past and future climate changes. Evaluating GCMs based on historical simulations is essential to improve credibility of the future climate projections. Previous studies found that CMIP5 models can better capture the observed characteristics of ENSO and its teleconnection influences including tropical and extra-tropical atmospheric circulations, compared to CMIP3 models (e.g., Kim and Yu 2012; Langenbrunner and Neelin 2013; Polade et al. 2013; Bellenger et al. 2014; Hurwitz et al. 2014; Roy et al. 2019). However, no study has evaluated the CMIP5 models in terms of ENSO influences on extreme SWHs.

This study conducts a novel analysis of the performances of 18 CMIP5 models for reproducing the observed ENSO influence on extreme SWH over the Indo-Pacific region during 1950–2005 boreal winters. We use a non-stationary generalized extreme value (GEV) distribution analysis (Kharin and Zwiers 2005; Kumar et al. 2016; Patra et al. 2020) to obtain the extreme SWH response patterns to ENSO from each model and compare the results with that from the ERA-20C reanalysis. For the two selected hotspots having strong responses to ENSO, we explore the factors responsible for the model skills in simulating the regional extreme SWH responses to ENSO using multi-model samples. In particular, the role and relative importance of the ENSO intensity and its teleconnection patterns in determining model skills are examined for the first time. Further, SWH responses to ENSO are compared with sea level pressure (SLP) gradient responses to reaffirm physical mechanisms associated with model performances. This paper is structured as follows. Section 2 describes the data (both models and reanalyses) and methodology used, including the extreme value analysis and the definition of skill metrics. CMIP5 model evaluation results for the Indo-Pacific Ocean and the two hotspots are given in Sect. 3, followed by the discussion in Sect. 4. Conclusions are provided in Sect. 5.

2 Data and methodology

2.1 SWH and SLP data

We used DJF mean (Havg) and maximum (Hmax) SWHs for the period 1950–2005 from 18 CMIP5 models (Table 1), which were calculated from 6-hourly wave height data downscaled from their historical simulations (Wang et al. 2014) and are downloadable from the Open Government Canada website (<https://open.canada.ca/data/en/dataset/d82b1c57-e766-495a-8a81-3bc1ebba87ae>). The 6-hourly SWHs were constructed from sea level pressure (SLP) using a statistical modeling approach (Wang et al. 2012), which is based on the relationship between atmospheric predictors such SLP and SLP

Table 1 18 CMIP5 models used in this study for generating SWH using a statistical downscaling method. See Sect. 2.1 for details

Model	Resolution	Institution
ACCESS1.0	145×192 L38	Commonwealth Scientific and Industrial Research Organization (CSIRO) and Bureau of Meteorology, Australia
BCC-CSM1-1	64×128 L26	Beijing Climate Centre, China Meteorological Administration
BCC-CSM1-1-m	160×320 L26	Beijing Climate Centre, China Meteorological Administration
CanESM2	64×128 L35	Canadian Centre for Climate Modelling and Analysis
CCSM4	192×288 L27	National Centre for Atmospheric Research, USA
CNRM-CM5	128×256 L31	Centre National de Recherche Meteorology, Meteo-France
CSIRO-Mk3-6-0	96×192 L18	Australian Commonwealth Scientific and Industrial Research Organization
GFDL-ESM2M	90×144 L24	Geophysical Fluid Dynamics Laboratory, USA
HadGEM2-ES	145×192 L40	UK Met Office Hadley Centre
INMCM4	120×180 L21	Institute for Numerical Mathematics, Russia
IPSL-CM5A-MR	143×144 L39	Institute Pierre-Simon Laplace, France
MIROC5	128×256 L40	Model for Interdisciplinary Research on Climate, Japan
MIROC-ESM	64×128 L80	Model for Interdisciplinary Research on Climate, Japan
MIROC-ESM-CHEM	64×128 L80	Model for Interdisciplinary Research on Climate, Japan
MPI-ESM-LR	96×192 L47	Max Planck Institute for Meteorology, Germany
MPI-ESM-MR	96×192 L95	Max Planck Institute for Meteorology, Germany
MRI-CGCM3	160×320 L48	Meteorological Research Institute, Japan
NorESM1-M	96×144 L26	Norwegian Climate Centre

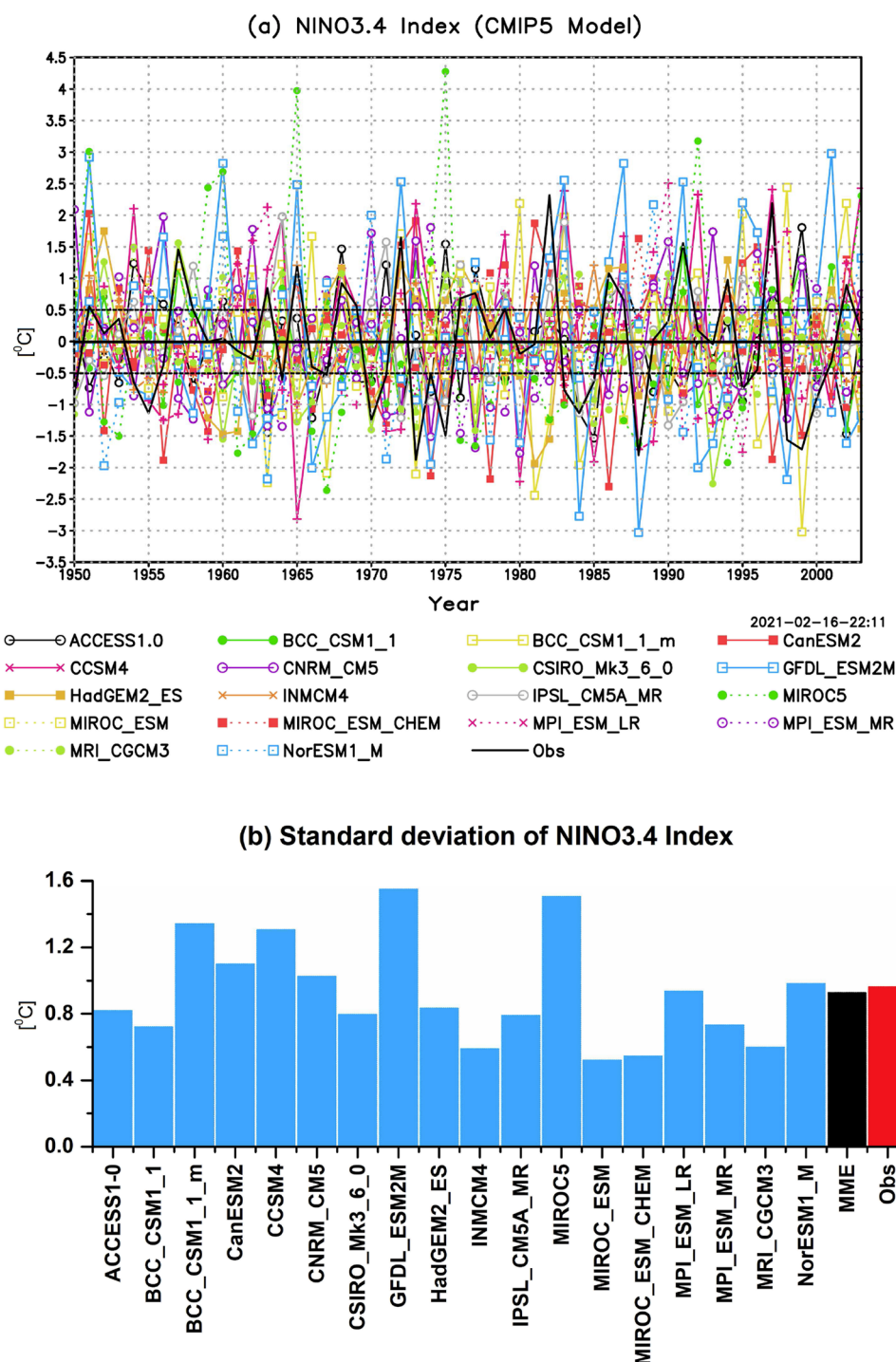
gradient (SLPG) and the wave variables like SWH. The statistical wave model was calibrated using the 6-hourly SWH and SLP data from the ERA-Interim reanalysis (Dee et al. 2011), as described in Wang et al. (2012). DJF mean and maximum SLP gradients were also calculated from 6-hourly SLP data of the CMIP5 models to examine the relationship between SLPG and SWH. For each CMIP5 model, single simulation was used and all data were interpolated onto a common $1^\circ \times 1^\circ$ grid before analysis (see Table 1 for their original resolutions). Note that although SWH data are not directly generated by CMIP5 models, we use CMIP5 model names for simplicity when interpreting their downscaled SWH results below.

As observations, we used 6-hourly SWH and SLP data from ERA-20C (Poli et al. 2013) during 1950–2005, which have a $1^\circ \times 1^\circ$ resolution and downloadable from the ECMWF website (<http://apps.ecmwf.int/datasets>), and obtained Havg and Hmax and DJF mean and maximum SLPG. To assess uncertainty associated with different reanalysis datasets, we also used ERA5 reanalysis which provides high-resolution ($0.5^\circ \times 0.5^\circ$) ocean wave outputs but covers a short period since 1979 (Hersbach et al. 2020), obtained from the website (<https://www.ecmwf.int/en/forecasts/datasets/reanalysis-datasets/era5>). For better comparison with CMIP5 model outputs, the calculated seasonal mean and extreme SWH and SLPG were interpolated into the same $1^\circ \times 1^\circ$ resolution.

2.2 ENSO indices

As an observed ENSO index, we used the Niño3.4 index (defined as SST anomalies averaged over 5°S – 5°N , 170°W – 120°W) during 1950–2005 calculated from the Hadley Centre Global Sea Ice and Sea Surface Temperature dataset (HadISST) version 2.1.0.0 (Titchner and Rayner 2014), which were used as the boundary condition for ERA-20C. The Niño3.4 indices for CMIP5 models (Table 1) were calculated for the same period using each model simulation's SSTs. The observed and simulated detrended DJF mean Niño-3.4 index time series are displayed in Fig. 1, together with corresponding inter-annual standard deviations of Niño3.4 indices. All models possess year-to-year fluctuations of Niño3.4 index, amplitudes of which vary across models. BCC-CSM2-1-m, CCSM4, GFDL-ESM2M, and MIROC5 exhibit larger ENSO amplitudes ($> 1.2^\circ \text{C}$) than the observed (0.97°C). CNRM-CM5, MPI-ESML-R, and NorESM1-M show similar variabilities to the observations while several models simulate weaker ENSO. Multi-model ensemble mean (MME) exhibit an ENSO amplitude (0.93°C) comparable to the observed. By linking the larger inter-model spread in ENSO amplitudes to those in SWH responses, we explore the possible influence of ENSO intensity on the simulated SWH responses to ENSO (see below).

Fig. 1 **a** Time series of DJF mean detrended Niño3.4 index during 1950–2005 and **b** inter-annual standard deviation of Niño3.4 index for eighteen CMIP5 models and HadISST v2.1 (Obs). Horizontal dashed lines in **a** represent ± 0.5 standard deviation from Obs. Multi-model mean (MME) of standard deviations is also depicted in **b** for comparison



2.3 Extreme value analysis

We use an extreme value analysis to assess extreme SWH responses to ENSO following previous studies (Kharin and Zwiers 2005; Kumar et al. 2016; Patra et al. 2020). Seasonal

maxima of 6-hourly SWH (Hmax) is fitted to a non-stationary Generalized Extreme Value (GEV) distribution on each grid, in which GEV parameters are basically set to vary with time (t). The cumulative density function of a non-stationary GEV distribution is given as

$$F(x; \mu_t, \sigma_t, \xi_t) = \begin{cases} \exp \left[-\exp \left(-\frac{x - \mu_t}{\sigma_t} \right) \right] & \text{for } \xi_t = 0 \\ \exp \left[-\left(1 + \xi_t \frac{x - \mu_t}{\sigma_t} \right)^{-\xi_t^{-1}} \right] & \text{for } \xi_t \neq 0 \text{ and } 1 + \xi_t \frac{x - \mu_t}{\sigma_t} > 0, \end{cases} \quad (1)$$

where μ_t , σ_t , and ξ_t denotes the location, scale, and shape parameters, respectively. The location parameter represents near-center position of the GEV distribution, corresponding to the mean intensity of extreme values, the scale parameter demonstrates the width or spread of the GEV distribution, describing inter-annual variability, and the shape parameter determines the tail behavior of the distribution (Coles 2001). Here, we introduce the climate variability ν_t (here detrended and normalized ENSO index) as a covariate of GEV location parameter as follows:

$$\mu_t = \mu_0 + \mu_1 (\nu_t - \nu_0) \quad (2)$$

where μ_0 denotes the location parameter at time t_0 and μ_1 denotes the regression coefficient, representing the influence of climate variability on extreme values (here ENSO influence on extreme SWH). We detrend the ENSO index to remove the possible influence of climate change induced long-term trends and then normalize it to consider different ENSO intensity among models, i.e. examining wave climate response to the same one standard deviation of ENSO variability across models.

A statistical significance in the influence of ENSO on extremes is assessed using the likelihood ratio test (Kharin and Zwiers 2005), which compares the log likelihood for the non-stationary GEV model (M_1) with that for the stationary GEV model (M_0). The log likelihood ratio statistic is defined as $D = 2(l^1 - l^0)$ where l^1 and l^0 are the likelihood of the model M_1 and M_0 , respectively. D follows a χ^2 distribution with degrees of freedom $q = 1$ (i.e. difference in the number of free parameters) and a statistically significant impact of ENSO is obtained if D is larger than the critical D value at 5% level. According to the likelihood ratio test, influence of ENSO on scale and shape parameters are found negligible. This is consistent with many previous studies (Kumar et al. 2016; Patra et al. 2020), implying that the inter-annual variability of extreme SWH and the tail behavior of GEV distribution remain largely unaffected by ENSO. We show results based on the non-stationary GEV model based on location parameter in Eq. (2) while keeping scale and shape parameters constant. Spatial patterns of GEV regression coefficients (μ_1) are obtained for individual CMIP5 models and compared with the ERA-20C pattern.

2.4 Model evaluation methods

To evaluate the overall spatial pattern of each model's extreme SWH responses to ENSO, we use a Taylor diagram skill score (S) that is calculated as follows (Taylor 2001):

$$S = \frac{4(1 + R)}{\left(\tilde{\sigma}_f + \frac{1}{\tilde{\sigma}_f} \right)^2 (1 + R_o)} \quad \text{with } \tilde{\sigma}_f = \frac{\sigma_f}{\sigma_o} \quad (3)$$

where σ_f , σ_o , R , and R_o denotes the (spatial) standard deviation of model, standard deviation of observation, correlation coefficient between model and observation, and maximum correlation attainable, respectively. The skill score quantifies how closely the simulated pattern resembles the observed pattern considering both spatial variability and spatial correlation. We also obtain linear regression coefficients of seasonal mean SWH and SLP onto Niño 3.4 index for each model and compare them with the observed pattern using the Taylor skill score.

To evaluate model-simulated SWH responses for the two selected hotspots (northeast NP and Maritime Continent; see below), area mean regression coefficients from models are compared with the observed. To further identify which factors (ENSO intensity or SLP teleconnection pattern) affect more importantly model performances at simulating regional SWH responses to ENSO, an inter-model correlation analysis is performed between model SWH responses and ENSO metrics as well as SLPG and SWH responses.

3 Results

3.1 Model evaluation for the Indo-Pacific region

We first evaluate models in terms of their SLP responses to ENSO over an Indo-Pacific Ocean region during DJF. Figure 2a displays the spatial patterns of regression coefficients of DJF mean SLP onto Niño3.4 index for individual CMIP5 models and multi-model mean (MME) in comparison with that obtained from ERA-20C reanalysis. The hatching represents regions with statistically significance at 5% level except for MME regression (regions with more than 80%

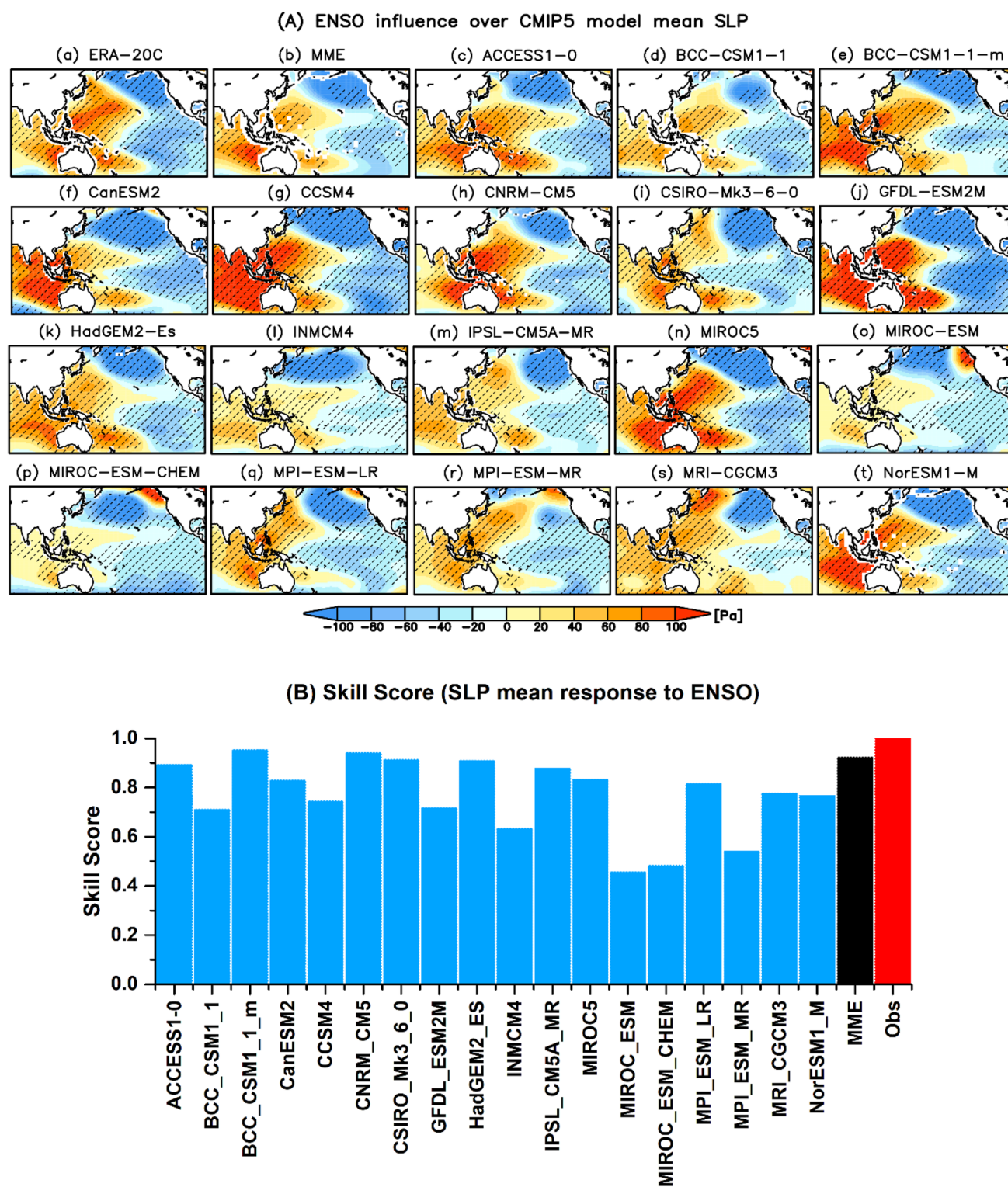


Fig. 2 **A** Spatial distributions of the regression coefficients of DJF SLP anomalies onto the Niño3.4 index during 1950–2005 from ERA-20C, 18 CMIP5 models and MME. The dashed hatching represents regions with significant regression at 5% level for ERA-20C and

individual CMIP5 models and regions with more than 80% model agreement in regression sign for MME. **B** Taylor skill scores of each CMIP5 model and MME for the SLP regression pattern

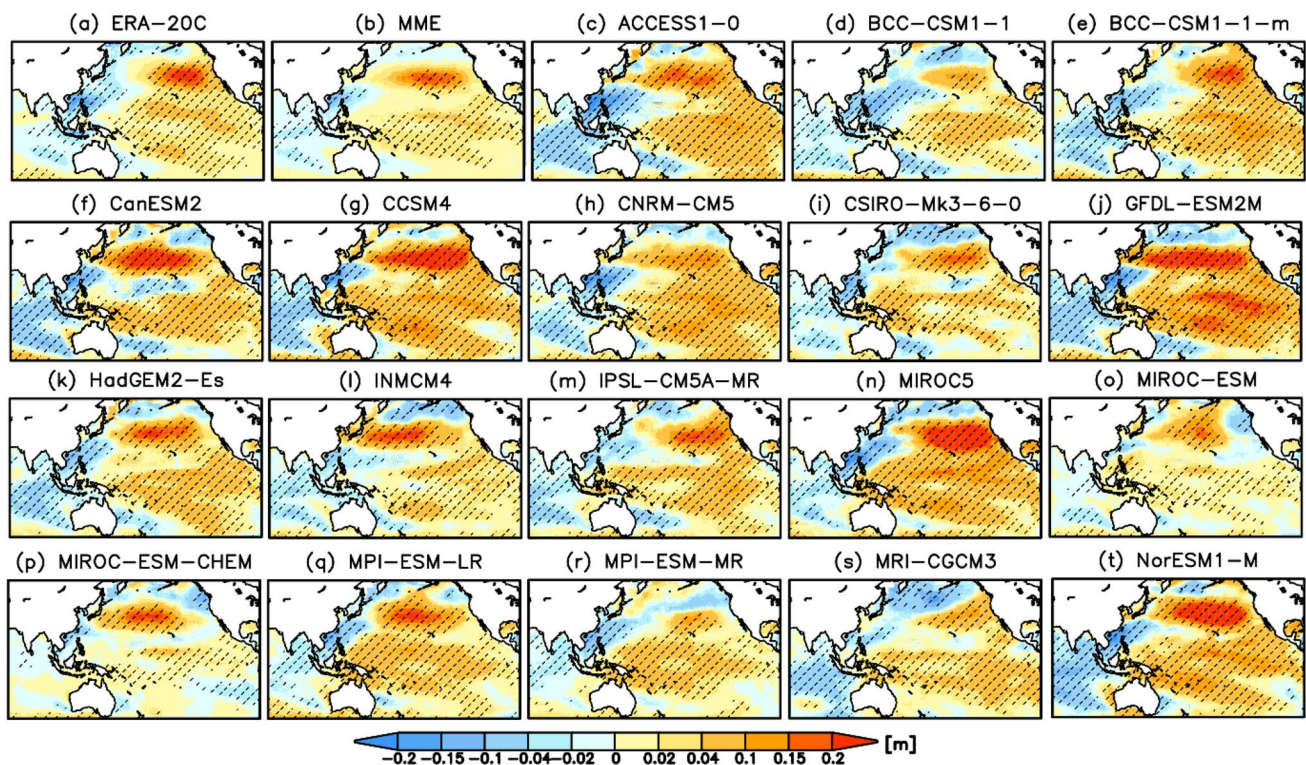


Fig. 3 Spatial distributions of the regression coefficients of DJF mean SWH (Havg) onto the Niño3.4 index during 1950–2005 from ERA-20C, 18 CMIP5 models and MME. Note that simple linear regression is used here, differently from a GEV-based regression for DJF max

SWH (Hmax). The dashed hatching represents regions with significant regression at 5% level for ERA-20C and individual CMIP5 models (based on the *t*-test) and regions with more than 80% model agreement in regression sign for MME

model agreement in the sign of regression coefficient). In ERA-20C, a low pressure anomaly is observed in eastern North Pacific (NP) and a high pressure anomaly over the western tropical Pacific and Indian Ocean area, which are captured reasonably by MME and many individual models. Some models (CCSM4 and GFDL-ESM2M) overestimate the observed dipole pattern while underestimation occurs by several other models (MIROC-ESM, MIROC-ESM-CHEM, MPI-ESM-MR, and INMCM4). Taylor skill scores (Fig. 2b) indicate that BCC-CSM1-1-m, CNRM-CM5, CSIRO-Mk3-6-0, and HadGEM2-ES can reproduce the observed SLP responses to ENSO with similar skills to MME. In contrast, models with weaker SLP responses (MIROC-ESM, MIROC-ESM-CHEM, MPI-ESM-MR, and INMCM4) exhibit lower skill scores. We use this Taylor skill score for SLP as each model's performance metric for ENSO teleconnection pattern to explore its influence on model skills for SWH responses.

Before examining extreme SWH (Hmax) responses, model skills for DJF mean SWH (Havg) responses to ENSO are evaluated. Figure 3 shows the spatial patterns of regression coefficients of Havg onto Niño3.4 for ERA-20C, MME, and individual models. The ERA-20C pattern is characterized by the increase in Havg over the northeastern NP

(NENP) and the decrease in Havg over the Maritime Continent (MC) during El Niño (Fig. 3a). Havg increases are also observed along the Intertropical convergence zone (ITCZ) and South Pacific convergence zone (SPCZ), consistent with previous studies (Patra et al. 2020). Most of the CMIP5 models can capture the observed Havg responses, including the positive response in NENP and negative response in MC, as can be seen in the MME pattern (Fig. 3b). There are, however, large variations in regression patterns (amplitude and location) among different models. For the Havg increase over NENP, some models overestimate the observed response with a westward shift (CanESM2, CCSM4, GFDL-ESM2M, MIROC5, and NorESM1-M) while other models underestimate it (BCC-CSM1-1, MPI-ESM-MR, and MRI-CGCM3). For the negative response in MC, overall weaker SWH responses are simulated by models (Fig. 3b) and, particularly, MIROC-ESM and MIROC-ESM-CHEM exhibit almost no response.

The observed Hmax response to ENSO remarkably resembles the Havg pattern but with stronger amplitudes (Fig. 4). This includes an increase over NENP and a decrease in MC and moderate increases along ITCZ and SPCZ during El Niño, suggesting identical physical mechanisms at work between extreme and mean SWHs (Kumar et al.

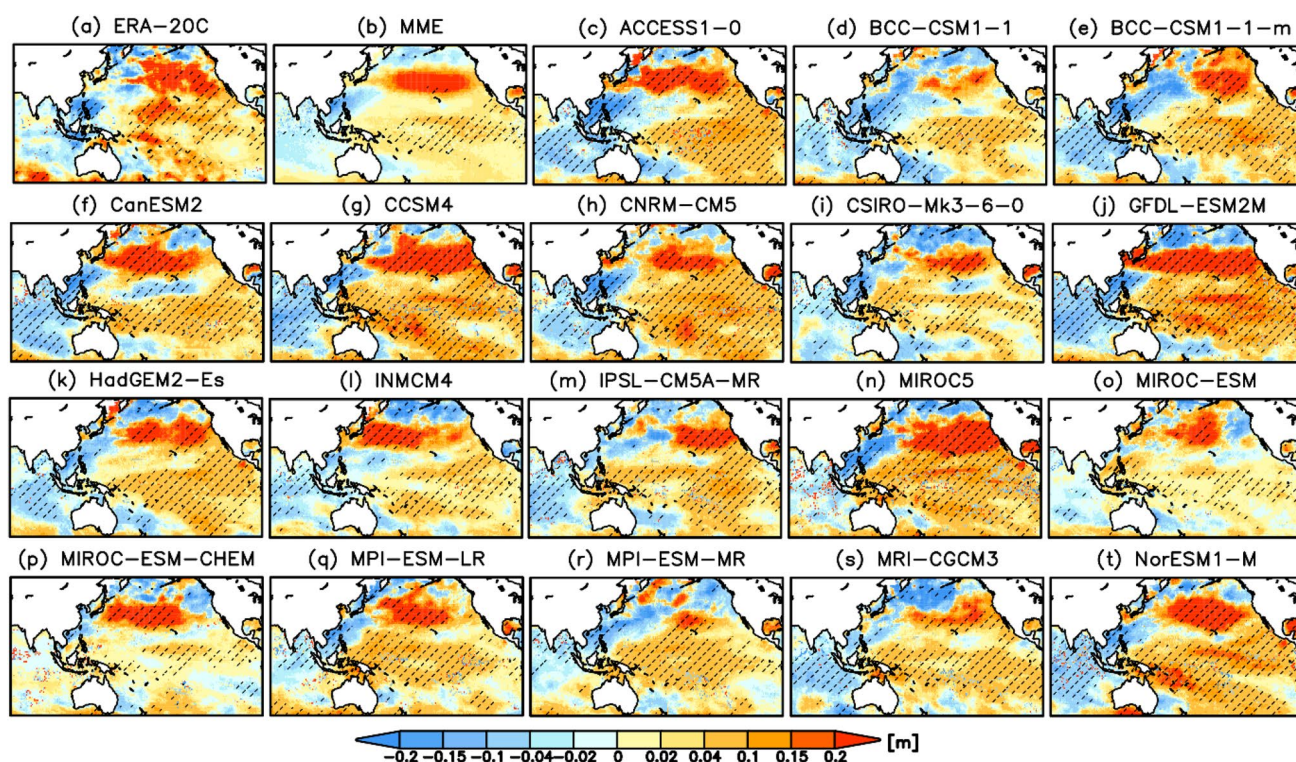


Fig. 4 Spatial distributions of the regression coefficients of DJF maximum SWH (Hmax) onto the Niño3.4 index during 1950–2005 from ERA-20C, 18 CMIP5 models and MME. Note that GEV location parameter is used here, which represents mean intensity of Hmax [μ_1 in

Eq. (2)]. The dashed hatching denotes regions with significant regression at 5% level (based on a likelihood ratio test) for ERA-20C and individual CMIP5 models and regions with more than 80% model agreement in regression sign for MME

2016; Patra et al. 2020). For the NENP region, models with stronger mean responses tend to simulate stronger extreme responses (CanESM2, CCSM4, GFDL-ESM2M, MIROC5, and NorESM1-M) and models with weaker mean responses show weaker extreme SWH responses (BCC-CSM1-1, MPI-ESM-MR, and MRI-CGCM3). The same occurs for the negative response in MC with overall underestimation by many models. Spatial correlation between the mean and extreme SWH response is very high ($r=0.92$), confirming their close association.

Results from Taylor diagram analysis quantify the model skills for the spatial pattern similarity of mean and extreme SWH responses to ENSO (Fig. 5a, b). Overall, skills for Havg are better than those for Hmax, as indicated by higher spatial correlation, and MME exhibits better skills than individual models for both Havg and Hmax due to the bias cancelation among models. As for the spatial variability, the diagrams for Havg and Hmax clearly indicate model underestimation with smaller standard deviation than ERA-20C. Four models with relatively larger standard deviation are well identified (CCSM4, GFDL-ESM2M, MIROC5, and NorESM1-M), which correspond to those with overestimated SWH responses as analyzed above. Models located closer to the ERA-20C reference point (or perfect model)

can be regarded as better skilled models, which include ACCESS10-0, BCC-CSM1-1-m, and CNRM-CM5. Resulting Taylor skill scores confirm these evaluation results (Fig. 5c, d). It can be seen that overall skills are higher for Havg than Hmax and the inter-model spread in Taylor skill scores is also larger for Havg (0.44–0.93) than Hmax (0.53–0.73). The latter can be seen in the Taylor diagram (Fig. 5a, b) such that models are more scattered in Havg than Hmax.

3.2 Model evaluation for the two hotspot regions

To explore what factors determine model skills on SWH responses to ENSO, two hotspot regions are selected as NENP (175° E–145° W, 25° N–46° N) and MC (100° E–133° E, 5° S–20° N) based on the observed response strength of extreme SWH to ENSO. The NENP and MC are regions with major ENSO influences, with significant increases and decreases in extreme SWH, respectively, during El Niño events (opposite responses during La Niña). Previous studies based on other reanalyses and satellite observations (Kumar et al. 2016; Patra et al. 2020) support that these are areas experiencing robust ENSO influences.

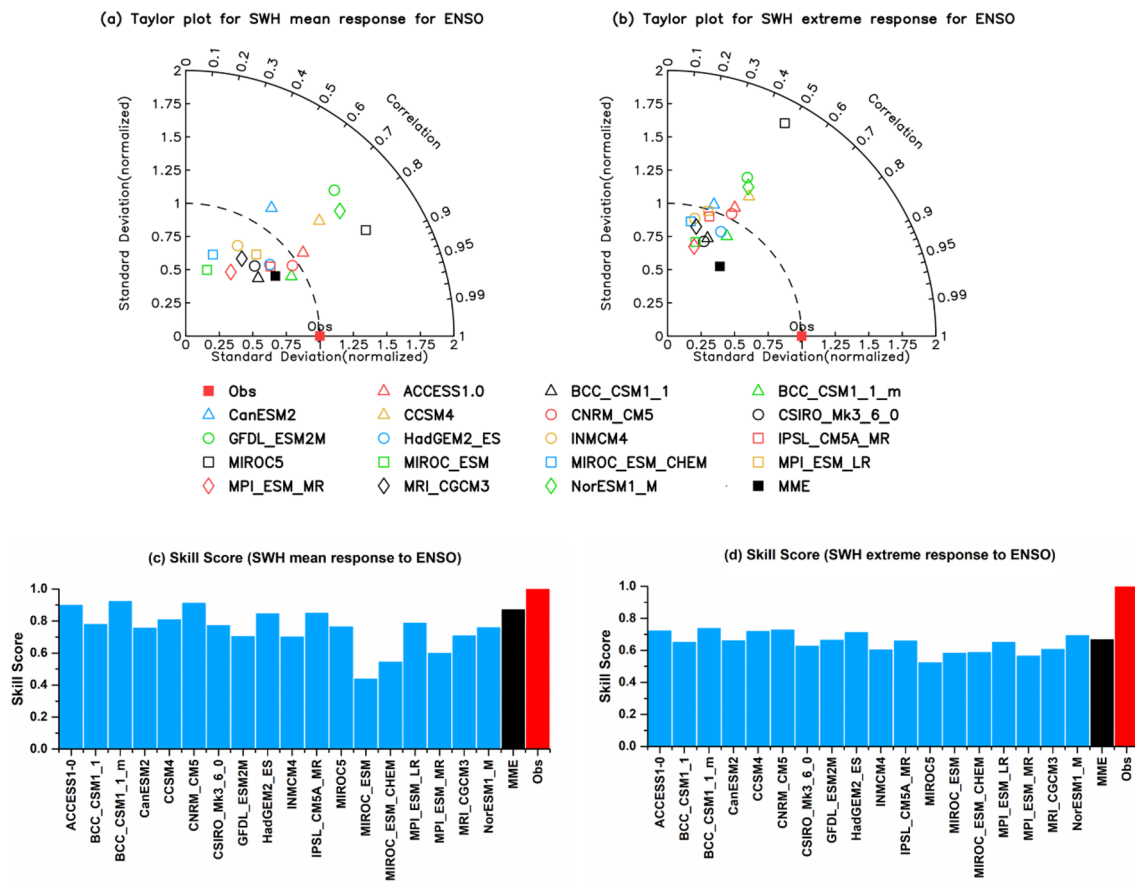


Fig. 5 Taylor diagram for **a** Havg and **b** Hmax responses to ENSO for 18 CMIP5 models and MME and their corresponding skill scores (**c**, **d**)

For each hotspot region, we examine relationships between ENSO amplitudes and SWH responses and also between Taylor skill scores for SLP and SWH responses using an inter-model correlation analysis based on 18 CMIP5 model values. For the NENP region (Fig. 6), a very strong correlation ($r=0.95$) is obtained between the area mean regression coefficients for Havg and Hmax, which is significant at 1% level (see Fig. 6b). This reflects that ENSO influence on Hmax are closely related to ENSO influence on Havg over the region. When compared with the observed value, many models have weaker responses for both variables. The four models (CCSM4, GFDL-ESM2M, MIROC5, and NorESM1-1) that oversimulate ENSO influences are identified clearly. A significant high inter-model correlation (>0.7) is obtained between the ENSO amplitude and Hmax/Havg responses to ENSO (Fig. 6c, d), which represents that the ENSO intensity is a driving factor explaining the large inter-model difference in Hmax/Havg responses to ENSO over the NENP. Indeed, models with stronger ENSO (such as GFDL-ESM2M, MIROC5, and CCSM4) tend to have stronger SWH responses while models with weaker ENSO (such as MIROC-ESM, MIROC-ESM-CHEM, INMCM4, and MRI-CGCM3) simulate weaker SWH responses to

ENSO. In contrast, insignificant and weak positive correlations ($r=0.34/0.27$) are found between Taylor skill scores for SLP and Havg/Hmax responses to ENSO (Fig. 6e, f). This implies that model's overall skill for SLP teleconnection pattern over the Indo-Pacific domain does not affect much the regional SWH responses over the NENP. The Taylor skill score measures the similarity of each model's SLP regression pattern onto its Niño3.4 index to the observed pattern (Fig. 2). However, it considers the entire Indo-Pacific domain, thus can be limited in representing regional SLP responses such as the NENP. In this respect, when checking the direct association between regional SLP and SWH responses, a significant inter-model correlation is observed (see below).

A very strong relation ($r=0.96$) between the Havg and Hmax responses is obtained for the MC region (Fig. 7b). Compared to the NENP case, model underestimation is much stronger with MME averaged regression coefficients being about a half of the observed. Only a few models (CNRM-CM5, BCC-CSM1-1, and MIROC5) can reproduce the observed magnitude of Hmax/Havg responses to ENSO (within $\pm 30\%$ of the observed values). An overestimated response occurs in ACCESS1-0. Also unlike the

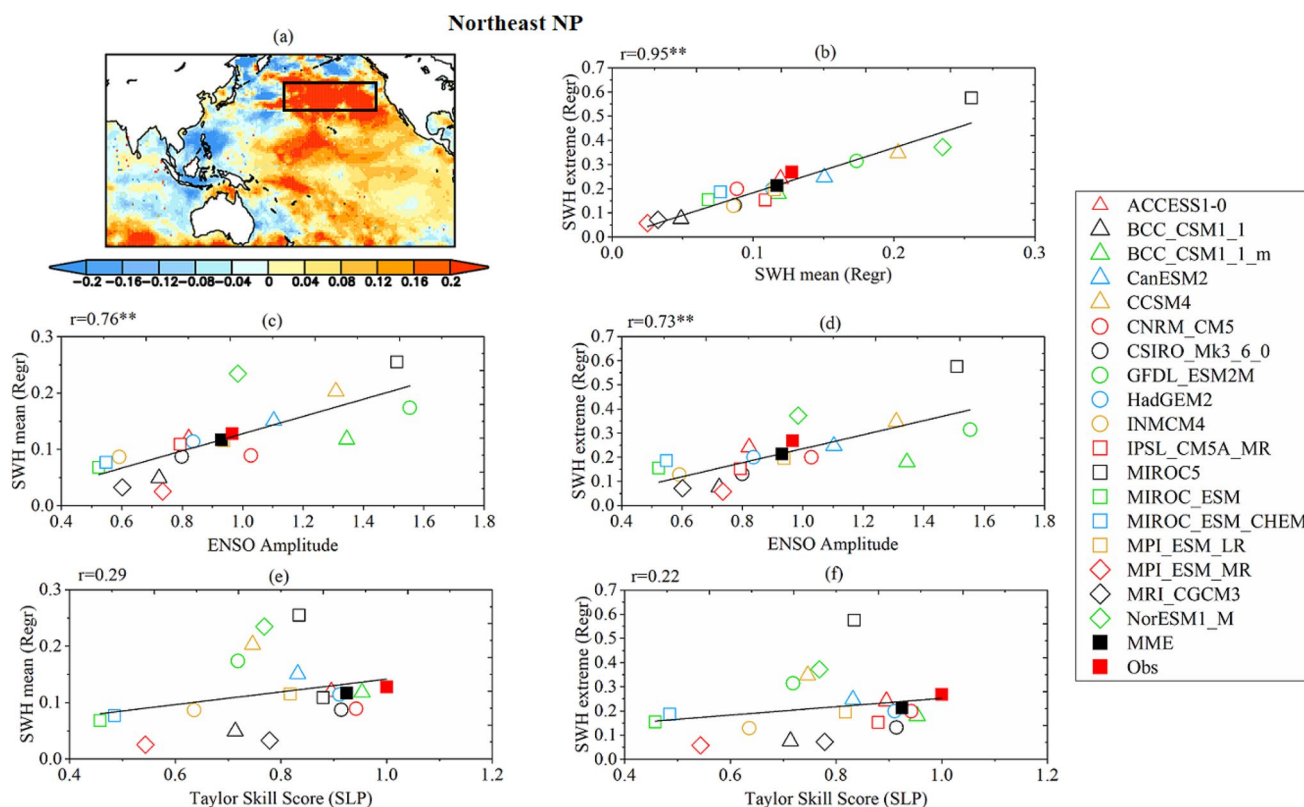


Fig. 6 **a** Same as Fig. 4a but for the Northeast NP (NENP, 175° E–45° W, 25° N–46° N) region indicated, **b–f** scatter plots and linear regression lines (solid black lines) representing the relationship between **b** Havg response and Hmax response to ENSO, **c** ENSO amplitude and Havg response, **d** ENSO amplitude and Hmax response, **e** Taylor skill score for SLP response and Havg response,

and **f** Taylor skill score for SLP response and Hmax response across 18 CMIP5 models. MME and ERA-20C points are also indicated for comparison. Inter-model correlation coefficients (r) are given at the top left. Single and double asterisks indicate statistically significant correlation at 5% and 1% level, respectively

NESM results, a significant negative correlation (< -0.5) is seen between the Taylor skill score of SLP and Havg/Hmax responses to ENSO for the MC region (Fig. 7e, f). Instead, ENSO amplitude is found to have a weak impact with a moderate negative correlation (< -0.3). This indicates that the SLP teleconnection patterns are more important for simulating Havg/Hmax responses to ENSO over the tropical western Pacific region where the weakening of the Walker circulation associated with the eastward shift of convection zone plays a critical role during El Niño (Klein et al. 1999; Wang 2002; Yeh et al. 2018; Bayr et al. 2020).

As a way of checking how regional wind changes induce SWH responses, DJF mean and extreme SLP gradient (SLPG) responses to ENSO are calculated for the two hotspot regions and compared with Havg/Hmax responses (Fig. 8). A strong inter-model association between Havg and mean SLPG response to ENSO (Fig. 8a, c) is obtained for the NENP and MC regions ($r=0.79$ and 0.86 , respectively). Hmax and extreme SLPG responses also show close relations ($r=0.81$ for NENP; $r=0.64$ for MC), indicating the importance of SLP-driven wind changes in determining SWH responses. When comparing with the ERA-20C results, models

can simulate the observed SLPG increase during El Niño for the NENP. In contrast, most of the CMIP5 models underestimate mean/extreme SLPG responses to ENSO for the MC region, explaining overall underestimation of SWH responses. This seems to be related to the underestimated Walker circulation responses by CMIP5 models (Bayr et al. 2020; Wang et al. 2021), which can be also affected by equatorial Pacific precipitation and SST biases (e.g., Langenbrunner and Neelin 2013; Bellenger et al. 2014). Nevertheless, our results support that extreme ocean wave responses to ENSO are determined by regional-scale atmospheric circulation changes. As a whole, we find the important role of ENSO intensity and SLP teleconnection patterns in shaping SWH responses to ENSO and they need to be taken into account with care when evaluating and improving models.

4 Discussion

In this section, we assess dependency of our results on ENSO index and reanalysis data and also discuss remaining issues. First, one may ask if there will be any

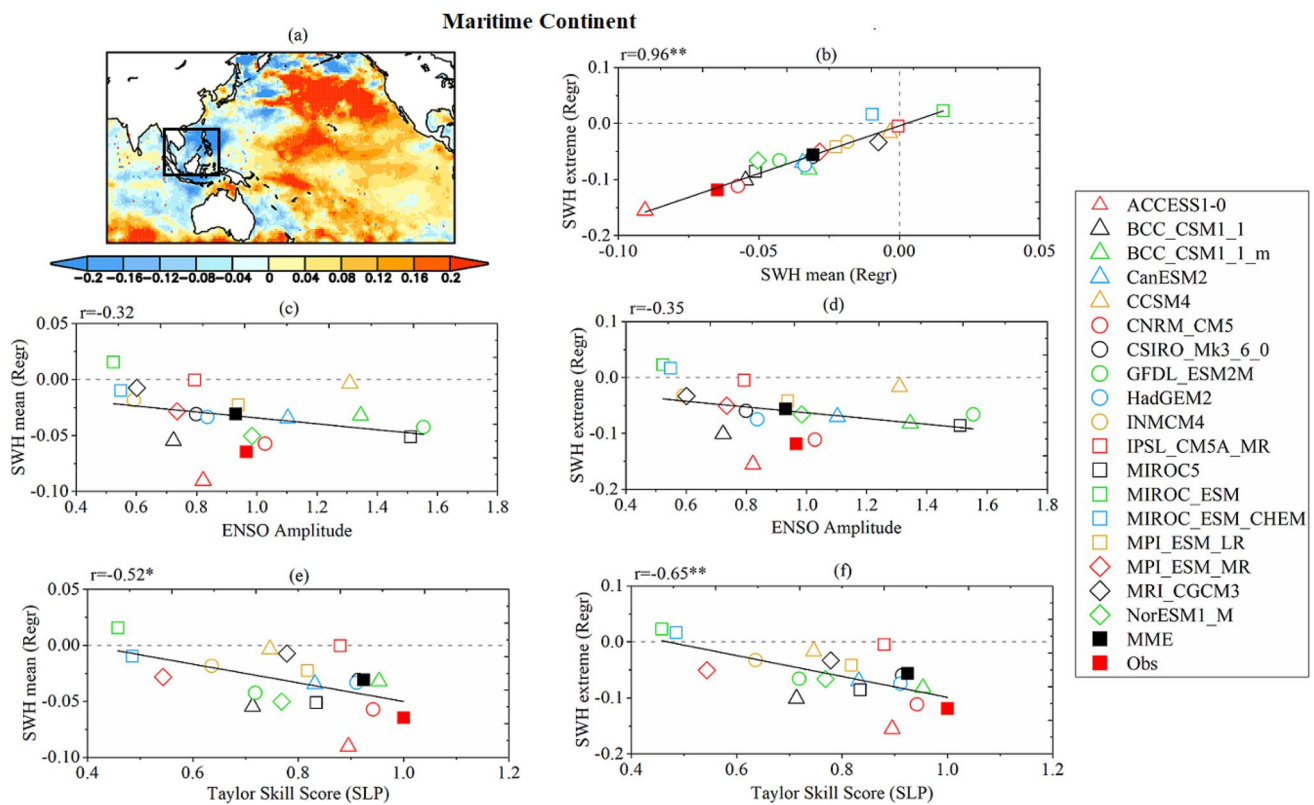


Fig. 7 Same as Fig. 6 but for the Maritime Continent (MC, 100° E–133° E, 5° S–20° N) region

differences if other ENSO indices are used such as the multivariate ENSO index (MEI, Wolter and Timlin 2011), which provides a more complete description of ENSO by additionally considering atmospheric circulation and convection responses. However, MEI values cannot be computed from CMIP5 models due to data limitations, for which the five variables of SLP, SST, zonal and meridional components of the surface wind, and outgoing long-wave radiation are required. Instead, we have compared the MEI and Niño3.4 index time series in the observations for 1950–2005 DJFs. A very high correlation between them ($r=0.978$) indicates that our results will remain almost the same if MEI is applied. Although Niño3.4 index is used in this study, which covers the overall ENSO domain of central-to-eastern equatorial Pacific, Niño3 and Niño4 indices or their ratios can be used in future work to evaluate models for different types of El Niño, i.e. eastern Pacific and central Pacific El Niño events (Kao and Yu 2009; Kug et al. 2009; Yeh et al. 2018).

Secondly, we have tested sensitivity of our results to the use of ERA5 reanalysis. Although it provides high resolution ($0.5^\circ \times 0.5^\circ$) ocean wave data, ERA5 is available for a shorter period since 1979 and relatively limited for examining ENSO influences and evaluating associated model skills. Therefore, we have briefly compared ERA5-based regression

patterns with those from ERA-20C for the same 1979–2005 period. Response patterns of SLP, Havg, and Hmax remain very similar between the two reanalyses (Fig. 9), representing negligible influence of different input observations and resolution employed between the two reanalyses. ERA-20C shows consistent regression patterns with the longer period case, indicating the weak influence of different analysis period as well. Also see our discussion on the resolution influence below.

Thirdly, caveats remain associated with the statistical downscaling method used in this study. In this respect, our results need to be further assessed using those based on dynamic downscaling, which is able to better produce the physical response to GCM winds and also provide additional wave parameters like period and direction as well as height. However, in dynamic downscaling, biases in model-simulated predictors (6-hrly wind) can induce large biases in the estimated predictand (wave parameters), as evaluated in Wang et al. (2010). Therefore, the statistical downscaling approach generally shows better performance than dynamical projections in reproducing the observed climatology and variability, by reducing the climate model bias through calibration to the observations. In this study, climate model simulated SWH and SLP data were adjusted to have the same climatology and inter-annual variability as those of

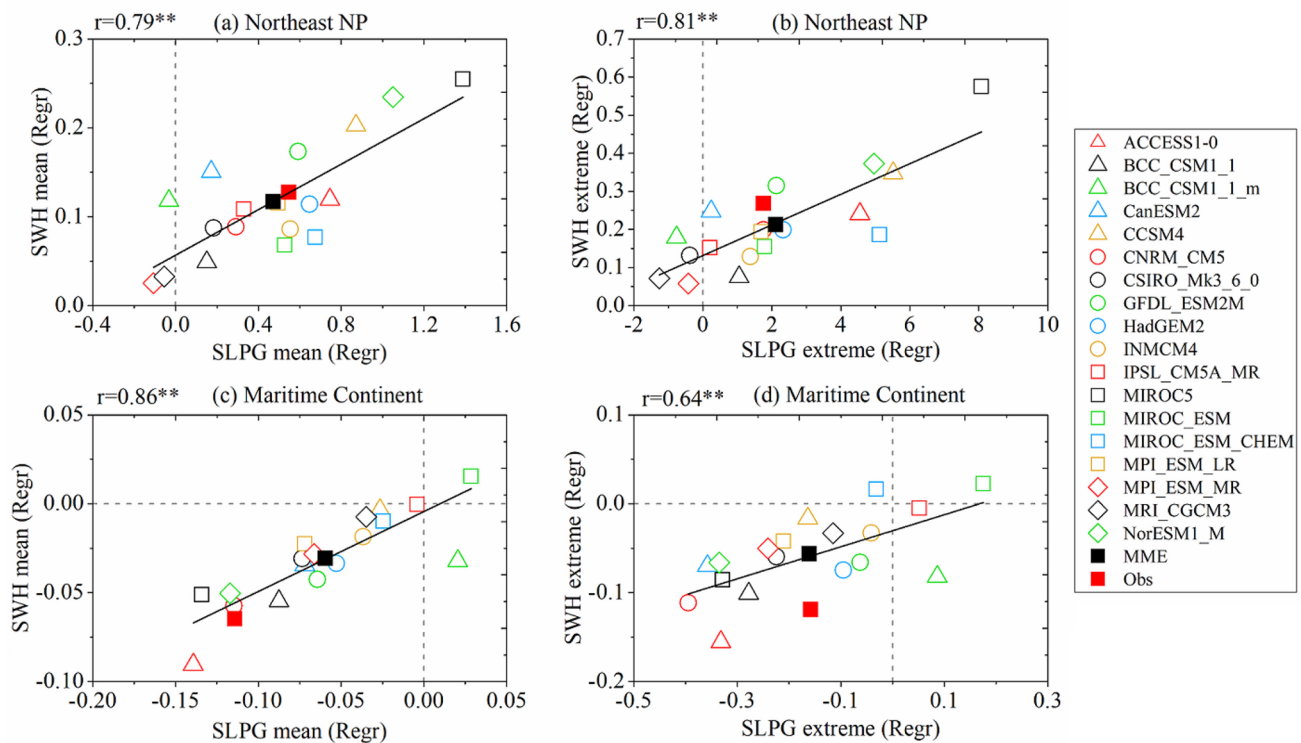


Fig. 8 Scatter plots and linear regression lines (solid black lines) representing the relationship between DJF mean SLP gradient (SLPG) response and Havg/Hmax responses averaged over **a**, **b** NENP and **c**,

d MC region. The statistically significant correlations at 1% and 5% level are marked with double and single asterisks, respectively

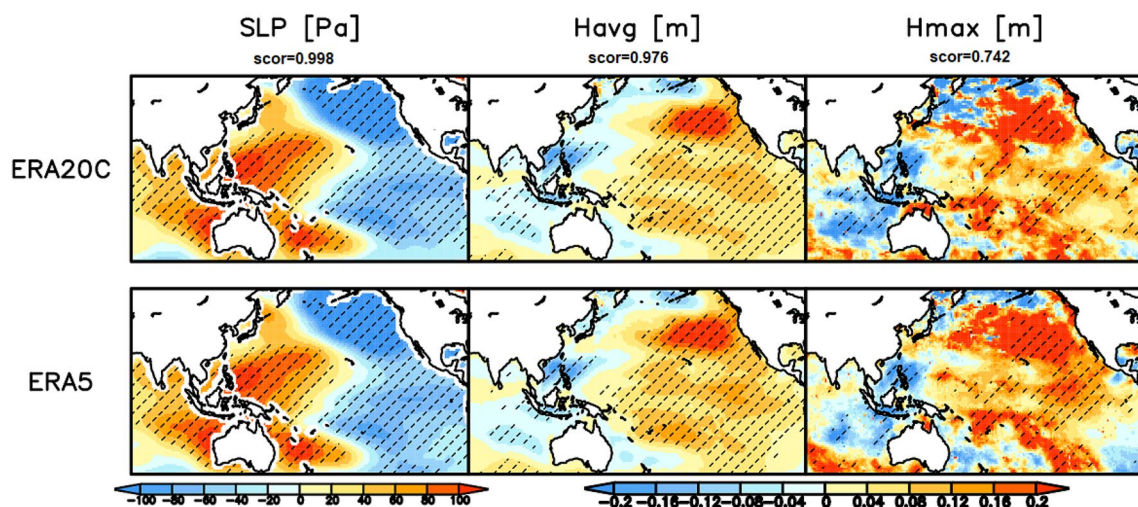


Fig. 9 Same as Figs. 2a, 3a, and 4a but for using (upper) ERA-20C and (lower) ERA5 based on the common 1979–2005 period. Spatial correlation values (scor) between regression patterns for each variable are provided

ERA-Interim reanalysis, as described in Wang et al. (2014). In addition, the dynamical downscaling methods require a high-quality bathymetry and wind fields and are

computationally more expensive than the statistical downscaling methods.

Lastly, this study considers DJF response only when ENSO has the strongest intensity and therefore its influence

on extreme wave heights is stronger, affecting larger areas, than other seasons (Patra et al. 2020). From the same reason, most previous studies evaluated climate models for ENSO characteristics and its teleconnection patterns, focusing on DJF. However, ENSO can affect boreal summer (June–August, JJA) wave heights through tropical cyclone (TC)-related changes over the western North Pacific (WNP, Yang and Oh 2018). It is well understood that TC genesis locations are shifted eastward and equatorward during El Niño, which makes TC survive longer and favors their movement toward WNP (Camargo and Sobel 2005; Chen and Lian 2018). Patra et al. (2020) also found significant increases in JJA wave heights over the WNP during El Niño from the satellite data and ERA5 reanalysis. However, CMIP5 models have significant deficiency in the TC formation and tracks due to their coarse resolutions (Camargo 2013; Emanuel 2013) and high-resolution model simulations are required to explore this important issue in a separate study (e.g., Yang and Oh 2020).

5 Conclusions

This study assesses the performance of CMIP5 models in simulating DJF mean and maximum SWH (Havg and Hmax) responses to ENSO over the Indo-Pacific Ocean and its subregions. 18 CMIP5 models are evaluated by comparing their historical simulations with ERA-20C reanalysis for 1950–2005. A non-stationary GEV analysis is employed to fit Hmax and identify regions with significant impacts of ENSO by using Niño3.4 index as a covariate of GEV location parameter. Results show that CMIP5 models can reproduce the observed Havg and Hmax responses to ENSO reasonably well, in particular, the increases over the north-eastern North Pacific (NENP) and the decreases over the Maritime Continent (MC) during El Niño and vice versa. CMIP5 model skills are further examined for the selected two hotspot regions by analyzing inter-model relationship between ENSO characteristics and SWH responses. While over the NENP region, ENSO intensity is found to play a crucial role in inducing Havg/Hmax responses, the large-scale SLP teleconnection turns out to be more important for simulating Havg/Hmax responses to ENSO over the MC region. It is also shown that the inter-model spread in Havg/Hmax responses is closely related to the regional SLP gradient, indicating the critical role of wind-driven wave responses. Our results provide an important implication that improving extreme SWH variability simulations requires better representation of ENSO intensity as well as atmospheric teleconnection patterns. Further investigation is warranted for understanding the detailed physical processes of how different ENSO can induce different SWH responses.

In particular, the role of the intensity and location of tropical convection and the associated modeling parameters need to be identified.

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Declarations

Conflict of interest The authors declare no competing financial interests.

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